

	FIR VEHICLE TESTING		22/08/2025
FIR No :FIR/Testing-Outdoor- UA/520100/2025/00741			
Part No/ Group :52010160R			
Subject :Performance evaluation of Signa 3522.T Cx RDE CNG AC in comparison with Ashok Leyland 3520.T BSVI and Signa 3523.T BSVI RDE base vehicle.			
Test Vehicle	Project Name :Signa 3522.T Cx RDE CNG AC Model Name : Signa 3522.T Cx RDE CNG AC		Enclosures : Annexure 1 - V
	Chasis Number :PC053604042RL1028 Engine Number :HVX106918		
FIR Status :	Completed		

Role	Name	Signature
Prepared By	Debabrata Das	 22-Aug-2025
Checked By	Swapnil Jadhav	 22-Aug-2025
Approved By	Ajaykumar Jadhav	 22-Aug-2025

Cc: Head CVBU Engg, Chief Engineer, Head COC, Project Team [Design]; Project Team [Testing]

PA20 F222 (R4) March, 2021

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Documents : Please click here

[FIR_Testing_Outdoor_UA_52..](#)

 FIR/Testing-Outdoor- UA/520100/2025/00741	FIR VEHICLE TESTING	Date: 22/08/2025	
Part No / Group: 52010160R	Specification no :- " 259052070002 "	Action by	Failure class

Subject: -	Performance evaluation of Signa 3522.T Cx RDE CNG AC in comparison with Ashok Leyland 3520.T BSVI and Signa 3523.T BSVI RDE base vehicle.		
➤ <u>Conclusion:</u> ➤ <u>Signa 3522.T CNG RDE AC in comparison with Ashok Leyland 3520 BSVI</u> - Overall performance of Signa 3522.T BSVI RDE CNG AC is inferior as compared to Ashok Leyland 3520.T BSVI. PALS position achieved “C” against “C” as vehicle meeting PALS target. ➤ <u>Signa 3522.T CNG RDE AC in comparison with Signa 3523.T Cx BSVI RDE</u> - Overall performance of Signa 3522.T BSVI RDE CNG AC is inferior as compared to Signa 3523.T BSVI RDE base vehicle.	--	--	
1.1 <u>Acceleration:</u> ➤ <u>Signa 3522.T CNG RDE AC in comparison with Ashok Leyland 3520 BSVI</u> - Acceleration (0-60 kmph) of Signa 3522.T BSVI RDE CNG AC was found to be 65.49 Sec. Signa 3522.T BSVI RDE CNG AC is 5.66% inferior as compared to Ashok Leyland 3520.T BSVI vehicle but meets PALS target. - Acceleration (0-1000m) of Signa 3522.T BSVI RDE CNG AC was found to be 83.17 Sec. Signa 3522.T BSVI RDE CNG AC is 2.58% inferior as compared to Ashok Leyland 3520.T BSVI vehicle but meets PALS target. ➤ <u>Signa 3522.T CNG Cx RDE AC in comparison with Signa 3523.T BSVI RDE</u> - Acceleration (0-60 kmph) of Signa 3522.T BSVI RDE CNG AC is 9.97% inferior as compared to Signa 3523.T BSVI RDE base vehicle. - Acceleration (0-1000m) of Signa 3522.T BSVI RDE CNG AC is 2.97% inferior as compared to Signa 3523.T BSVI RDE base vehicle.	--	--	
1.2 <u>Wide open throttle drivability:</u> ➤ <u>Signa 3522.T CNG RDE AC in comparison with Ashok Leyland 3520 BSVI</u> In gear drivability of Signa 3522.T BSVI RDE CNG AC vehicle was found to be 2.26 kmph/s, 2.69 kmph/s, 2.08 kmph/s, 1.23 kmph/s, 0.94 kmph/s, and 0.63 kmph/s in 1st , 2nd , 3rd , 4th , 5th and 6th gear respectively.	--	--	
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 FIR/Testing-Outdoor- UA/520100/2025/00741	FIR VEHICLE TESTING	Date: 22/08/2025
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➤ Rate of acceleration of Signa 3522.T BSVI RDE CNG AC vehicle is superior to Ashok Leyland 3520.T BSVI in all gears. ➤ <u>Signa 3522.T CNG Cx RDE AC in comparison with Signa 3523.T Cx BSVI RDE</u> • Rate of acceleration of Signa 3522.T BSVI RDE CNG AC vehicle is superior to Signa 3523.T BSVI RDE CNG AC in all gears.	--	--
1.3 <u>Bathtub drivability:</u> ➤ <u>Signa 3522.T BSVI RDE CNG in comparison with Ashok Leyland 3520</u> • Bathtub drivability of Signa 3522.T BSVI RDE CNG AC vehicle was found inferior as compared to Ashok Leyland 3520.T BSVI vehicle in all gears. Vehicle meeting PALS target in all gears except 3rd gear (10 – 15 Kmph) and 4th Gear (25 – 30 kmph). ➤ <u>Signa 3522.T BSVI RDE CNG in comparison with Signa 3523.T BSVI RDE</u> • Bathtub drivability of Signa 3522.T BSVI RDE CNG AC vehicle was found inferior to Signa 3523.T BSVI RDE vehicle in all gears.	--	--
1.4 <u>Sustain and Restart Gradability:</u> ➤ <u>Signa 3522.T BSVI RDE CNG AC in comparison with Ashok Leyland 3520 BSVI</u> • Sustain gradeability of Signa 3522.T BSVI RDE CNG AC was found to be 22.1 %. Signa 3522.T BSVI RDE CNG AC is 2.08% inferior as compared to Ashok Leyland 3520.T BSVI vehicle but meeting PALS target. • Restart gradeability of Signa 3522.T BSVI RDE CNG AC was found to be 19.6%. Signa 3522.T BSVI RDE CNG AC is 5.1% superior as compared to Ashok Leyland 3520.T BSVI vehicle but meeting PALS target.	--	--
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2.0 <u>Result:</u> Wide observation.		

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FAILURES CLASS:

- A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement.
- B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention.
- C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule.
- D: Customer perception issues and minor irritants.

3.0 Test Vehicle:

1. Signa 3522.T BSVI RDE CNG AC(EL00313)
Chassis No. – PC053604042RL1028
Engine No. – HVX106918

Input Doc Ref. : Testing-Outdoor-UA/01941

DID No : NA

DDP No : NA

Enclosures:

- Annexure 1: Panel chart
Annexure 2: PAT Sheet
Annexure 3: PAT Summary
Annexure 4: Test request
Annexure 5: Test plan

Work Order No. :- P.21.EA.458

Make :- Tata Motors Ltd

T I No. – 001816

Earlier Ref: -

- FIR/Testing-Outdoor/520700/2023/6291_Performance evaluation on Signa 3523.T BSVI RDE vehicle.

4.0 Project Status: DR3

5.0 Introduction:

- As a part of RDE program, Signa 3522.T BSVI RDE CNG AC vehicle was offered for performance evaluation, Ashok Leyland 3520.T BSVI was taken as core benchmark and Signa 3523.T BSVI RDE vehicle was taken as Base for performance comparison. This report will reveal detail of this trial with the status.

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 FIR/Testing-Outdoor- UA/520100/2025/00741	FIR VEHICLE TESTING	Date: 22/08/2025	
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6.0 Testing

➤ Following test was conducted during trials.

Sr. No.	Test Description	Standard Referred	Instrument used
1	Acceleration	TST_VT_WG_0002	VBOX SX with GPS, Weighbridge (CRS Gate)
2	WOT Drivability	TST_VT_WG_0002	
3	Bath -Tub Drivability	TST_VT_WG_0261	
4	Gradability	TST_VT_WG_11	
5	Restart Gradability	TST_VT_WG_127	
6	Coastdown	TST_VT_WG_36	

7.0 Vehicle details:

Sr. No.	Parameters	Benchmark Vehicle BSVI	Base Vehicles BSVI RDE	Test Vehicle BSVI RDE
		Ashok Leyland 3520	Signa 3523.T RDE BSVI	Signa 3522.T RDE BSVI
1	Engine	6 Cylinder 5.7L / 200 HP/ BSVI	6 Cylinder 5.6 L / 230 HP BSVI	4 Cylinder 5.3 L / 220 HP BSVI
1.1	Engine Max. Power	200 hp @ 2400 RPM	220 HP	210 HP@2300 RPM
1.2	Engine Max. Torque	700 Nm @ 1200-2000 RPM	925Nm @ 1000-1300rpm	850 Nm @ 1100-1600 rpm
1.3	Power to weight (HP/Ton)	5.71	6.28	6.00
1.4	Torque to weight (Nm/Ton)	20	26.42	24.28
1.5	Engine Calibration Number			---
1.6	Engine Calibration Maturity		95%	95%
1.7	Emission Cal Maturity			---
1.8	Regeneration Strategy	Yes	Auto	NA
1.9	Urea Dosing	Yes	Yes	NA
2.0	Exhaust Downpipe Insulation		Updated downpipe (2.1)	NA
2.1	EOL Parameters	NA	LBSC - 2000,2000,35000 kg, VAM - 3.0,2.5, 10, 40 Km/h, GDP - 50, 50 Km/h, RSG - 80 Km/h	NA
2	Fuel Economy Improvement Technologies			
2.1	Engine Cooling Fan	E-Viscous	E-Viscous	E-Viscous
2.2	Fuel Economy Switch		Yes (3 Mode)	Yes (3 Mode)
2.3	Gear Shift Adviser		Yes	Yes
2.4	Accelerator Pedal Map		Yes	NA
3	Other Parameters			
3.1	Fuel Tank type	Polymer	Polymer	NA
3.2	Driveline	Eaton (6 speed), FGR 9.01	G 950 DD , FGR 9.36	G 950 DD , FGR 9.36
3.3	Gear Ratio	I-9.01, II-5.27, III-3.22, IV-2.04, V-1.36, VI-1	I-9.36, II-5.2, III-3.05, IV-1.84, V-1.33, VI-1,	I-9.36, II-5.2, III-3.05, IV-1.84, V-1.33, VI-1,
3.4	RAR	5.83	5.12	5.85
3.5	Tire size	Front – Rib 295/90R 20 Apollo Rear – Lug 295/90R20 Apollo	Front – Rib 295/90R 20 JK Tyre Rear – Lug 295/90R20 JK Tyre	Front – Rib 295/90R 20 Apollo Tyres Rear – Lug 295/90R20 Apollo Tyres

Vehicle Photographs



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Part No / Group: 52010160R	Specification no :- " 259052070002 "	Action by	Failure class

8.0 Torque table:-

➤ Signa 3522.T CNG Torque Table

3522.T 5.7L BSVI-II (CNG) <u>Heavy Mode</u>											
Heavy Mode		RAR	5.85								
G950	Gears	700	1000	1200	1400	1600	1800	2000	2200	2400	2750
9.36	1	295	800	850	850	850	810	740	685	635	0
5.2	2	295	800	850	850	850	810	740	685	635	0
3.05	3	295	800	830	830	830	810	740	685	635	0
1.84	4	295	800	830	830	830	810	740	685	635	0
1.33	5	295	800	830	830	830	810	740	685	635	0
1	6	295	800	830	830	830	810	740	685	635	0
3522.T 5.7L BSVI-II (CNG) <u>Medium Mode</u>											
G950 DD	Gears	700	1000	1200	1400	1600	1800	2000	2200	2400	2750
9.36	1	295	800	850	850	850	810	740	685	635	0
5.2	2	295	800	850	850	850	810	740	685	635	0
3.05	3	295	800	850	850	850	810	740	685	635	0
1.84	4	295	685	685	685	685	660	615	575	530	0
1.33	5	295	685	685	685	685	660	615	575	530	0
1	6	295	685	685	685	685	660	615	575	530	0
3522.T 5.7L BSVI-II (CNG) <u>Light Mode</u>											
G950 DD	Gears	700	1000	1200	1400	1600	1800	2000	2200	2400	2750
9.36	1	295	800	850	850	850	810	740	685	635	0
5.2	2	295	800	850	850	850	810	740	685	635	0
3.05	3	295	685	685	700	700	660	615	575	530	0
1.84	4	295	575	575	575	575	570	560	545	505	0
1.33	5	295	575	575	575	575	570	560	545	505	0
1	6	295	575	575	575	575	570	560	545	505	0

➤ 9.0 Observation:-

Overall performance Signa 3522.T BSVI RDE CNG AC is better as compared to Ashok Leyland 3520.T BSVI and PALS position achieved "C" against "C".

10. Future testing:- As per further development.

FIR Status

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PAT ATTRIBUTES SHEET

Performance and Drivability	Vehicle-level Engine Calibration
Features / End of Line	OBDII and Diagnostics

Document Version: VE CVBU/PD/00

Document Revision No.: VE CVBU/ M&HCV/ Signa 3522.T CNG Cx RDE
AC/ 00

Department Name: Vehicle Engineering

Project Name: Signa 3522.T CNG AC Cx BSVI RDE

Date: 22/08/2025

Aggregate and test parameter							Unit	Test Type	Classification (DICE)	Ref.Standard / Work Guideline	Approved	Base Vehicle	Core Benchmark	Target	Test Vehicle	Status (P/F/D)	Remarks	CoC Remark
Sr.No.	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7						Signa 2523.T 5.6L Aswamedh BSVI RDE G850 DD 6S RAR 5.12	Ashok Leyland 3520.T Eaton 5 Speed DD RAR 5.83		Signa 2523.T Cx RDE BSVI AC			
Performance Evaluation																		
A1	Minimum speed						Heading	Heavy Brake	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	
I.1.1	1st gear						Kmph		Specification		5.1	2.3	--	5.1	--			
I.1.2	2nd gear						Kmph		Specification		5.5	3.5	--	5.1	--			
I.1.3	3rd gear						Kmph		Specification		8.4	6.5	--	9.0	--			
I.1.4	4th gear						Kmph		Specification		10.6	10.6	--	14.2	--			
I.1.5	5th gear						Kmph		Specification		21.6	16.2	--	22.2	--			
I.1.6	6th gear						Kmph		Specification		28.7	23.3	--	28.0	--			
I.1.7	7th gear						Kmph		Specification		NA	NA	--	NA	--			
I.1.8	8th gear						Kmph		Specification		NA	NA	--	NA	--			
A2	Maximum speed						Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	
I.2.1	1st gear						Kmph		Specification		7.8	25.1	--	9.0	--			
I.2.2	2nd gear						Kmph		Specification		14.0	17.2	--	17.0	--			
I.2.3	3rd gear						Kmph		Specification		20.8	27.5	--	28.0	--			
I.2.4	4th gear						Kmph		Specification		30.5	43.5	--	47.5	--			
I.2.5	5th gear						Kmph		Specification		50.3	64.2	--	64.5	--			
I.2.6	6th gear						Kmph		Specification		80.3	80.0	--	79.4	--			
I.2.7	7th gear						Kmph		Specification		NA	NA	--	NA	--			
I.2.8	8th gear						Kmph		Specification		NA	NA	--	NA	--			
I.2.9	Vehicle maximum speed						Kmph		Specification		80.0	80	--	80	--			
A3	Acceleration						Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	
I.3.1	AC ON						Sec		Target		55.55	61.58	--	68.50	61.45			
I.3.2	0-60 kmph (0-37 mph)						Sec		Target		80.77	81.08	--	85.46	81.17			
I.3.3	Grad ability						Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	
I.3.4	AC ON						Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	
I.3.5	Maximum (sustained) Gradability (forward)						%		Standard		25.37	22.58	--	22.58	--			
I.3.6	Max Retard Gradability (forward)						%		Standard		28.8	18.6	--	216.68	18.6			
A4	Coast down (km/h²)						Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	
I.4.1	Laden condition						Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	
I.4.2	Power consumed at 40 kmph						kW		Data Generation		28.6	36.6	--	--	--			
I.4.3	a						%		Data Generation		1840.00	2259	--	--	--			
I.4.4	b						%		Data Generation		0.21	0.29	--	--	--			
I.4.5	Coeff of rolling resistance						N/m²/kgf		Data Generation		0.0001	0.0005	--	--	--			
Drivability Evaluation																		
B1	WOT Drivability (AC ON)						Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	
I.5.1	1st gear- 4-6 Kmph						Sec		Data Generation		5.07	5.11	--	5.09	--			
I.5.2	2nd gear- 8-12 Kmph						Sec		Data Generation		2.20	2.17	--	2.48	--			
I.5.3	3rd gear- 10- 20 Kmph						Sec		Data Generation		5.16	5.39	--	3.85	--			
I.5.4	4th gear- 10- 33 Kmph						Sec		Data Generation		14.65	13.66	--	11.38	--			
I.5.5	5th gear- 25- 42 Kmph						Sec		Data Generation		22.55	20.06	--	15.92	--			
I.5.6	6th gear- 36- 50 Kmph						Sec		Data Generation		27.86	22.98	--	23.28	--			
I.5.7	AC OFF (Rate of acceleration)						Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	
I.5.8	1st gear- 4-6 Kmph						Kmph/sec		Data Generation		1.86	1.81	--	2.26	--			
I.5.9	2nd gear- 8-12 Kmph						Kmph/sec		Data Generation		3.82	1.84	--	2.68	--			
I.5.10	3rd gear- 10- 20 Kmph						Kmph/sec		Data Generation		5.38	4.49	--	2.98	--			
I.5.11	4th gear- 10- 33 Kmph						Kmph/sec		Data Generation		0.96	1.11	--	1.23	--			
I.5.12	5th gear- 25- 40 Kmph						Kmph/sec		Data Generation		0.67	0.75	--	0.94	--			
I.5.13	6th gear- 36- 50 Kmph						Kmph/sec		Data Generation		0.64	0.43	--	0.64	--			
A5	Refinish Curve - Part Throttle Drivability (AC ON)						Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	
I.6.1	2nd gear						sec		Heading									
I.6.2	3rd gear						sec		Heading									
I.6.3	15-15 kmph						sec		Target/Balanced		4.51	2.96	--	4.33	--			
I.6.4	15- 20 kmph						sec		Target/Balanced		3.18	2.95	--	3.27	--			
I.6.5	4th gear						sec		Heading									
I.6.6	20-25 kmph						sec		Target/Balanced		6.20	4.13	--	6.45	--			
I.6.7	25-30 kmph						sec		Target/Balanced		3.94	3.63	--	4.41	--			
I.6.8	30-35 kmph						sec		Target/Balanced		4.09	4.20	--	4.33	--			
I.6.9	35-40 kmph						sec		Target/Balanced									
I.6.10	40-45 kmph						sec		Heading									
I.6.11	45-50 kmph						sec		Target/Balanced		12.29	12.85	--	13.27	--			
I.6.12	50-60 kmph						sec		Heading									
I.6.13	30- 40 kmph						sec		Target/Balanced		20.30	20.97	--	21.62	--			
I.6.14	40- 50 kmph						sec		Target/Balanced		16.88	20.15	--	22.80	--			
I.6.15	50- 60 kmph						sec		Target/Balanced		16.26	21.41	--	22.30	--			
Engine Calibration Activities																		
C1	System Verification Trials						Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	
I.7.1	Powertrain Characteristics (AC ON)						Target											
A6	Evaluation of Take-off ability, Acceleration and Deceleration						Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	Heading	
I.8.1	Take-off ability- Subjective Characteristics (**Please do Usability matrix to get the details of take and rolling values)						Target											
I.8.2	Test to measure Limp Home Engine speed that satisfies 7 deg Grade ability						Rating out of 10											
I.8.3	Test to measure Limp Home Engine speed that satisfies 7 deg Grade ability						Rating out of 10											
I.8.4	Test to verify vehicle performance affecting Engine parameters in dynamic condition						Rating out of 10											
I.8.5	Test to verify vehicle performance affecting Engine parameters in dynamic condition						Rating out of 10											

Sr.No.	Aggregate and test parameter						Unit	Test Type	Classification (I/O/R)	Ref.Standard / Work Guideline	Specification	Base Vehicle	Core Benchmark	Target	Test Vehicle	Status (RYG)	Remarks	CoC Remark							
	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7						Sigma 3523.T 5.4L, Aswameeth BSVI RDE 0850 DD 6S RAR 5.12	Ashok Leyland 3520.T Eaton 5 Speed DD RAR 5.83		Sigma 3522.T Cx RDE BSVI AC										
Sign Off															Index for Status										
Responsibility												Name							Signature			Off target, no resolution			
Prepared by: PAT Engineer												Debabrata Das/Swapnil Jadhav							 			Marginally off target, resolution planned/low risk			
Checked & Approved by: PAT Lead / Project Lead												Ajaykumar Jadhav / Sanjeev kumar							 			Met target, no concern			

PED evaluation of Signa 3522.T RDE BSVI CNG RDE in comparison with Ashok Leyland 3520 BSVI and Signa 3523.T Ashwamedh BSVI

PED| 21.08.25

OUR CULTURE CREDO

AT TATA MOTORS

We are connecting aspirations by being bold in thought and action, owning every opportunity and challenge, Solving together as one team and engaging all our stakeholders with empathy.

We are **MORE WHEN ONE!**

BE BOLD

Taking calculated **risk** is key to making progress. We act with confidence and **agility** to accomplish our goals

OWN IT

Feeling and acting **empowered** is critical to drive results. We have an **Owner's Mind-set** and each of us takes full responsibility for the outcomes

SOLVE TOGETHER

Leveraging our collective genius while holding each other **accountable** helps us deliver the best. We **collaborate** proactively and transparently to achieve innovative solutions

BE EMPATHETIC

Embracing diversity makes us stronger for differences are opportunities to learn. We work with **passion to delight customers** and deliver greater success to our stakeholders

- ✓ Vehicle specification comparison
- ✓ Performance Data Summary
- ✓ Minimum and Maximum Gear Speed
- ✓ Wide open throttle performance
- ✓ Tractive effort comparison
- ✓ Acceleration
- ✓ WOT drivability
- ✓ Bathtub drivability
- ✓ Coast Down Data
- ✓ Sustain and restart gradeability

Vehicle Specification Comparison

Sr. No.	Parameters	Benchmark Vehicle BSVI	Base Vehicles BSVI RDE	Test Vehicle BSVI RDE
		Ashok Leyland 3520	Signa 3523.T RDE BSVI	Signa 3522.T RDE BSVI
1	Engine	6 Cylinder 5.7L / 200 HP/ BSVI	6 Cylinder 5.6 L / 230 HP BSVI	4 Cylinder 5.3 L / 220 HP BSVI
1.1	Engine Max. Power	200 hp @ 2400 RPM	220 HP	210 HP@2300 RPM
1.2	Engine Max. Torque	700 Nm @ 1200-2000 RPM	925Nm @ 1000-1300rpm	850 Nm @ 1100-1600 rpm
1.3	Power to weight (HP/Ton)	5.71	6.28	6.00
1.4	Torque to weight (Nm/Ton)	20	26.42	24.28
1.5	Engine Calibration Number			---
1.6	Engine Calibration Maturity		95%	95%
1.7	Emission Cal Maturity			---
1.8	Regeneration Strategy	Yes	Auto	NA
1.9	Urea Dosing	Yes	Yes	NA
2.0	Exhaust Downpipe Insulation		Updated downpipe (2.1)	NA
2.1	EOL Parameters	NA	LBSC - 2000,2000,35000 kg, VAM - 3.0,2.5, 10, 40 Kmph, GDP - 50, 50 Kmph, RSG – 80 Kmph	NA
2	Fuel Economy Improvement Technologies			
2.1	Engine Cooling Fan	E-Viscous	E-Viscous	E-Viscous
2.2	Fuel Economy Switch		Yes (3 Mode)	Yes (3 Mode)
2.3	Gear Shift Adviser		Yes	Yes
2.4	Accelerator Pedal Map		Yes	NA
3	Other Parameters			
3.1	Fuel Tank type	Polymer	Polymer	NA
3.2	Driveline	Eaton (6 speed), FGR 9.01	G 950 DD , FGR 9.36	G 950 DD , FGR 9.36
3.3	Gear Ratio	I-9.01, II-5.27, III-3.22, IV-2.04, V-1.36, VI-1	I-9.36, II-5.2, III-3.05, IV-1.84, V-1.33, VI-1,	I-9.36, II-5.2, III-3.05, IV-1.84, V-1.33, VI-1,
3.4	RAR	5.83	5.12	5.85
3.5	Tire size	Front – Rib 295/90R 20 Apollo Rear – Lug 295/90R20 Apollo	Front – Rib 295/90R 20 JK Tyre Rear – Lug 295/90R20 JK Tyre	Front – Rib 295/90R 20 Apollo Tyres Rear – Lug 295/90R20 Apollo Tyres

Performance Data Summary

Parameter	Unit	Target	PALS	Benchmark	Base Vehicle			Test vehicle		RDE Status (%) w.r.t Benchmark	RDE Status (%) w.r.t Base	PALS status
				Ashok Leyland 3520 BSVI Eaton	Signa 3523.T BSVI RDE HM	Signa 3523.T BSVI RDE BM	Signa 3523.T BSVI RDE LM	Signa 3522.T RDE BSVI HM_AC	Signa 3522.T RDE BSVI HM_w/o_AC			
Acceleration at Rated Load												
0 - 60 kmph	sec	≤ 66.30	C	61.98	59.55	76.50	58.57 (0-50 kmph)	65.49	62.93	-5.66%	-9.97%	
0 - 1000 m	sec	≤ 85.46	C	81.08	80.77	86.50	89.29	83.17	80.91	-2.58%	-2.97%	
Driveability(Rate of acceleration) at Rated Load												
1st Gear(4-6)	sec	Not Applicable/ Data Generation	Not Applicable/ Data Generation	1.09	1.07	1.02	1.06	0.89	0.86	Not Applicable/Data Generation	Not Applicable/Data Generation	-
2nd Gear(8-12)	sec			2.22	2.20	2.20	2.35	1.49	1.35			
3rd Gear(12-20)	sec			4.44	5.76	5.88	5.97	3.85	3.61			
4th Gear(19-33)	sec			11.49	14.65	15.83	17.48	11.38	10.42			
5th Gear(25-40)	sec			20.30	22.55	26.43	29.86	15.92	15.50			
6th Gear(36-50)	sec			50.57	27.99	33.45	40.93	22.38	20.75			
1st Gear(4-6)	kmph/sec	Not Applicable/ Data Generation	Not Applicable/ Data Generation	1.84	1.86	1.97	1.89	2.25	2.33	Not Applicable/Data Generation	Not Applicable/Data Generation	-
2nd Gear(8-12)	kmph/sec			1.81	1.82	1.82	1.70	2.68	2.96			
3rd Gear(12-20)	kmph/sec			1.80	1.39	1.36	1.34	2.08	2.22			
4th Gear(19-33)	kmph/sec			1.22	0.96	0.88	0.80	1.23	1.34			
5th Gear(25-40)	kmph/sec			0.74	0.67	0.57	0.50	0.94	0.97			
6th Gear(36-50)	kmph/sec			0.48	0.50	0.42	0.34	0.63	0.67			
Bath-Tub Driveability												
3rd Gear												
10 --15	Sec	≤ 3.03	C	2.96	4.51	4.47	4.71	3.32	3.07	-12.16%	26.39%	
15 - 20	Sec	≤ 2.78	C	2.55	3.19	3.63	3.46	2.17	2.12	14.90%	31.97%	
4th Gear												
20 - 25	Sec	≤ 4.25	C	4.13	6.20	6.64	7.09	3.80	3.64	7.99%	38.71%	
25 - 30	Sec	≤ 4.01	C	3.63	3.94	4.42	4.57	4.16	3.67	-14.60%	-5.58%	
30 - 35	Sec	≤ 4.33	C	4.20	4.09	6.07	4.89	4.01	3.71	4.52%	1.96%	
35 - 40	Sec	--						4.81	4.61			
5th Gear												
30 - 40	Sec	≤ 13.27	C	12.85	12.29	12.70	19.50	11.19	10.64	12.90%	8.95%	
6th Gear												
30 - 40	Sec	≤ 21.62	C	20.97	29.30	32.42	32.60	18.84	15.41	10.15%	35.70%	
40 - 50	Sec	≤ 20.90	C	20.15	16.88	22.77	29.60	19.16	18.40	4.92%	-13.51%	
50 - 60	Sec	≤ 22.30	C	21.41	18.26			20.96	18.14	2.08%	-14.79%	
60 - 70	Sec	--										
Sustain and Restart Gradeability												
Sustained Gradeability	%	≥22.04	C	22.56	25.37	NA	NA	22.1		-2.08%	-12.89%	
Restart Gradeability	%	≥18.08	C	18.6	18.8	NA	NA	19.6		5.10%	4.26%	

Signa 3522.T Cx BSVI RDE Performance. PALS position achieved "C" against "UC"

Acceleration

- Acceleration of Signa 3522.T Cx BSVI RDE was found to be **inferior as compared to target** in 0-60 kmph speed and 0-1000 meter band

Bath tub Drivability

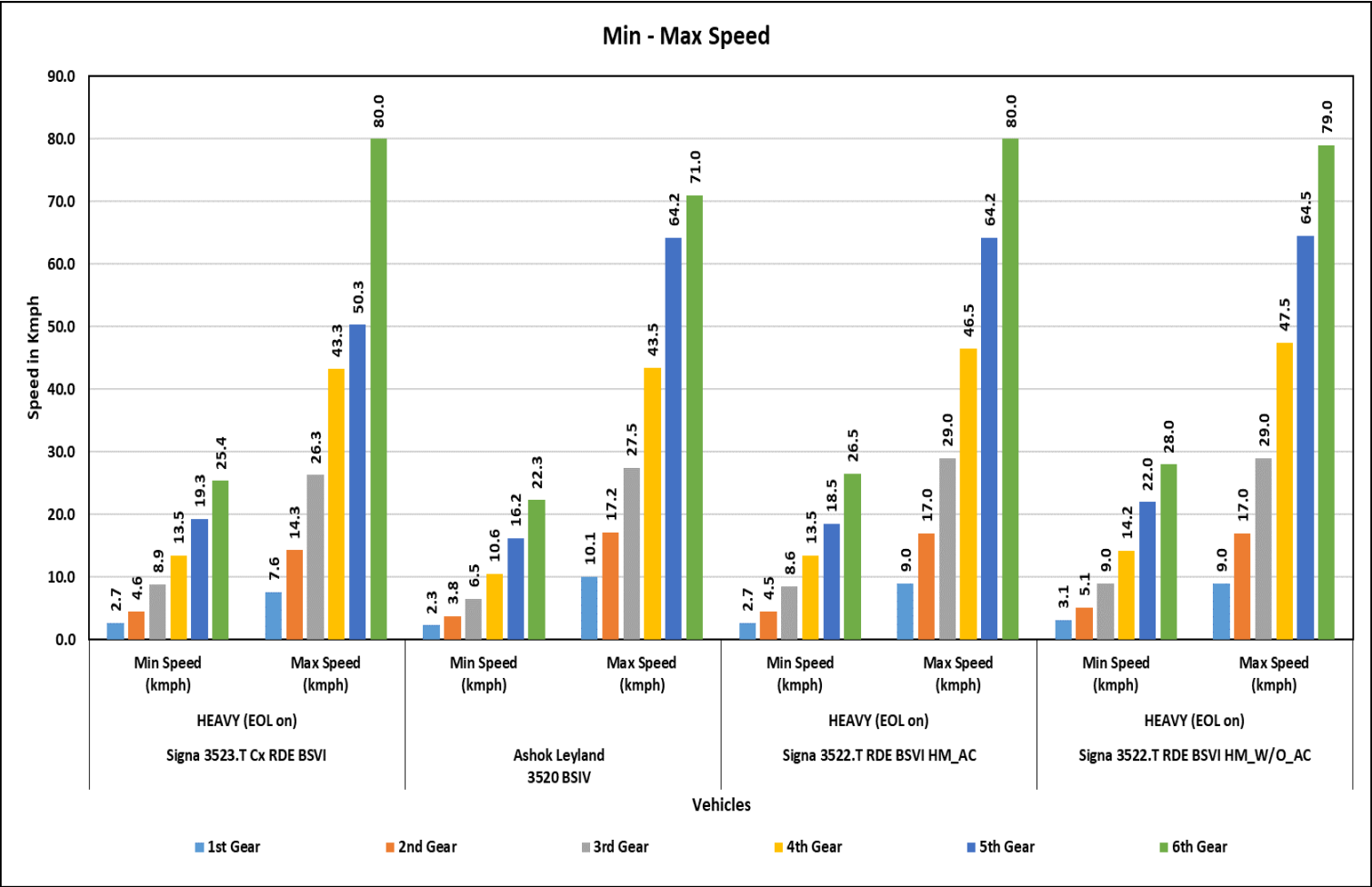
- Drivability of Signa 3522.T Cx BSVI RDE was found to be **better as compared to target** in 5th and 6th gear.
- Drivability of Signa 3522.T Cx BSVI RDE is **not meeting target** in 3rd Gear (10 – 15 Kmph) and 4th Gear (25 – 30 kmph)

Sustain & Restart Gradeability

- Restart Gradability of 3522.T Cx RDE BSVI was found better than target.
- Sustain Gradability of 3522.T Cx RDE BSVI was found better that target

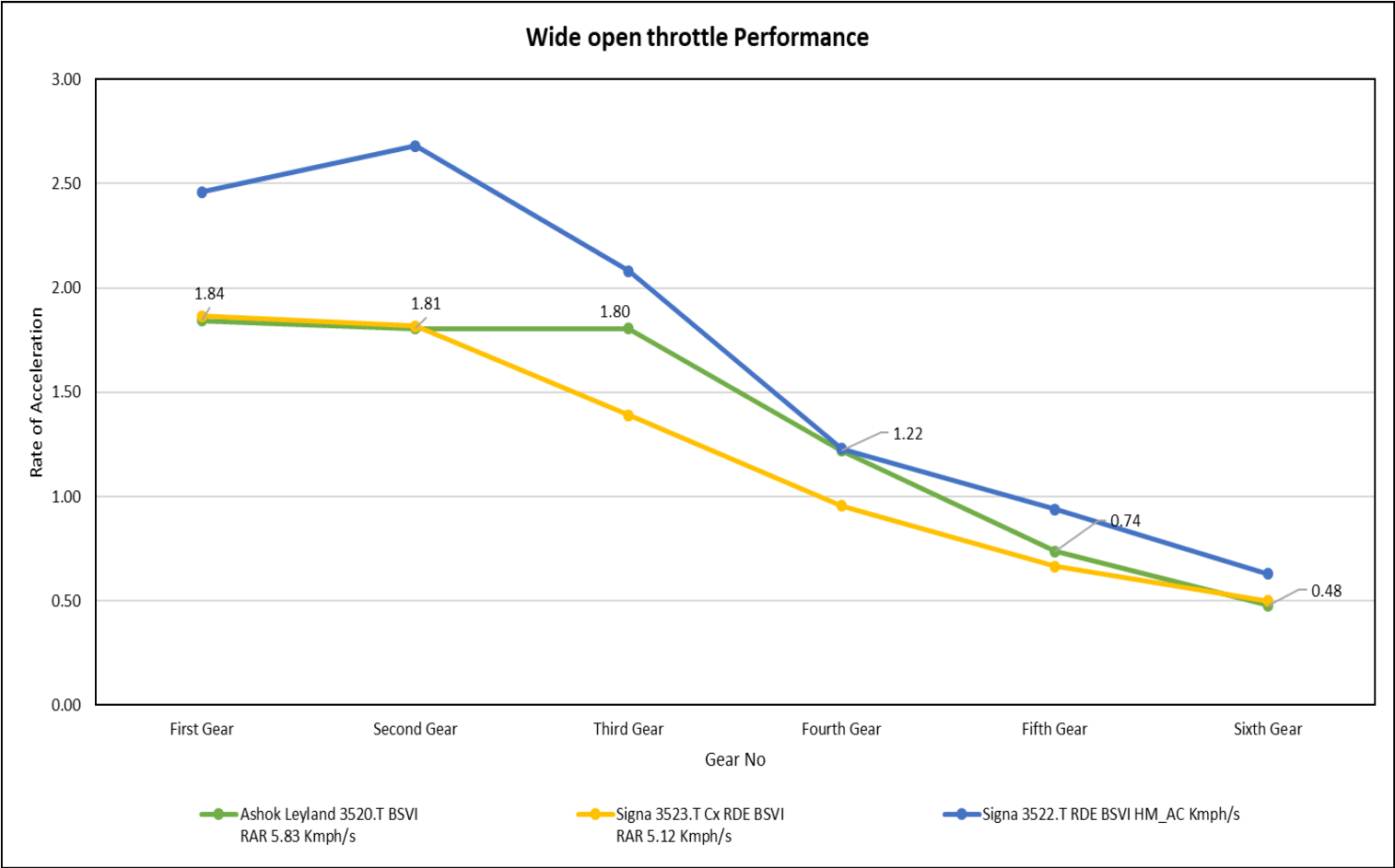
Minimum & Maximum Gear Wise speed Comparison

Test Vehicle	Base Vehicle		Benchmark Vehicle		Test Vehicle		Test Vehicle	
GEAR	Signa 3523.T Cx RDE BSVI		Ashok Leyland 3520 BSVI		Signa 3522.T RDE BSVI HM_AC		Signa 3522.T RDE BSVI HM_W/O_AC	
	HEAVY (EOL on)				HEAVY (EOL on)		HEAVY (EOL on)	
	Min Speed	Max Speed	Min Speed (kmph)	Max Speed (kmph)	Min Speed (kmph)	Max Speed (kmph)	Min Speed (kmph)	Max Speed (kmph)
1	2.7	7.6	2.3	10.1	2.7	9.0	3.1	9.0
2	4.6	14.3	3.8	17.2	4.5	17.0	5.1	17.0
3	8.9	26.3	6.5	27.5	8.6	29.0	9.0	29.0
4	13.5	43.3	10.6	43.5	13.5	46.5	14.2	47.5
5	19.3	50.3	16.2	64.2	18.5	64.2	22.0	64.5
6	25.4	80.0	22.3	71.0	26.5	80.0	28.0	79.0



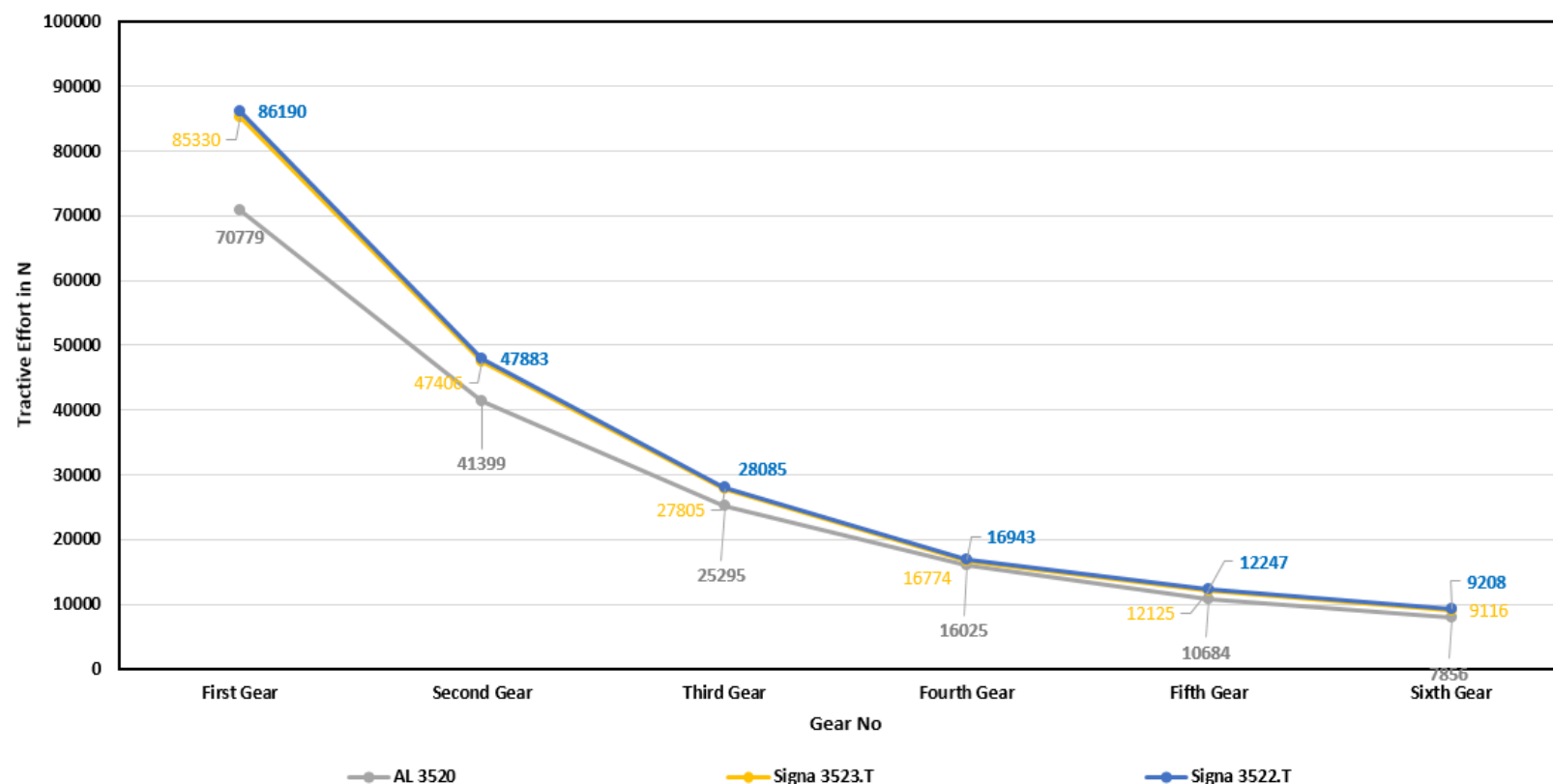
Wide open throttle Performance

Gear No	Ashok Leyland 3520.T BSVI RAR 5.83	Signa 3523.T Cx RDE BSVI RAR 5.12	Signa 3522.T RDE BSVI HM_AC
	Kmph/s	Kmph/s	Kmph/s
First Gear	1.84	1.86	2.46
Second Gear	1.81	1.82	2.68
Third Gear	1.80	1.39	2.08
Fourth Gear	1.22	0.96	1.23
Fifth Gear	0.74	0.67	0.94
Sixth Gear	0.48	0.50	0.63



Tractive effort comparison with Benchmark

Tractive Effort in N

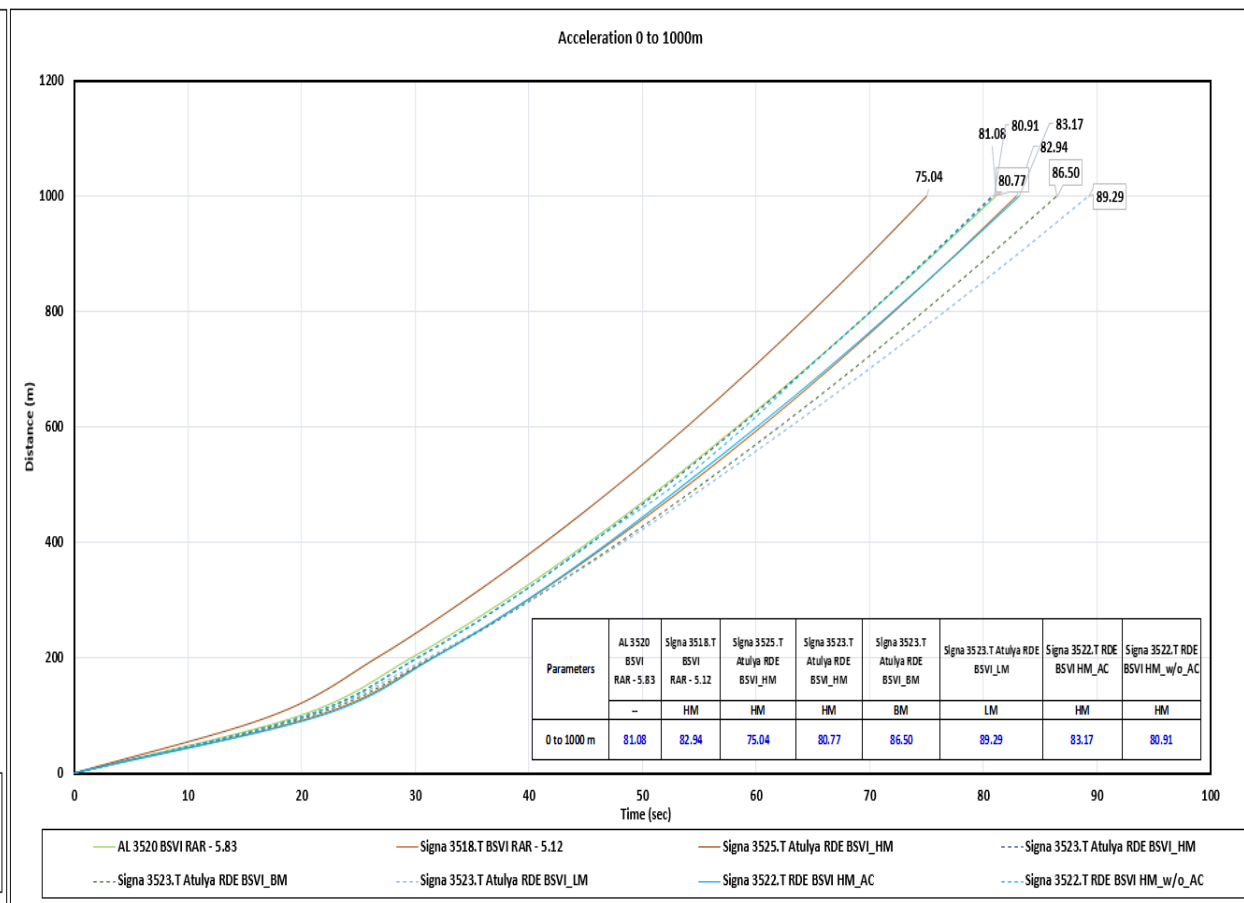
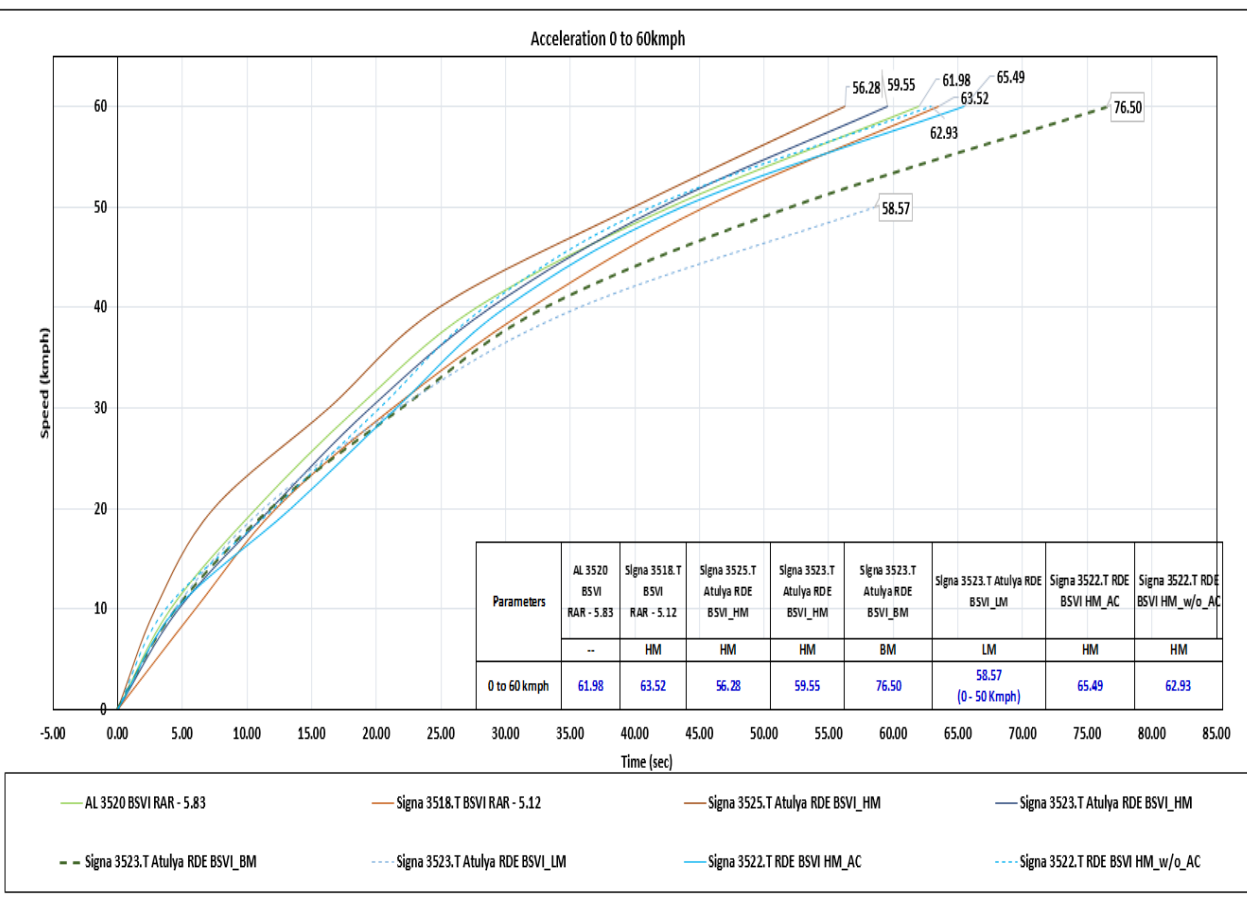


Remarks:

- Tractive effort of Signa 3522.T RDE BSVI is Superior than Benchmark & Base Vehicle vehicle in all gear.

Gear			
	AL 3520	Signa 3523.T	Signa 3522.T
First Gear	70779	85330	86190
Second Gear	41399	47406	47883
Third Gear	25295	27805	28085
Fourth Gear	16025	16774	16943
Fifth Gear	10684	12125	12247
Sixth Gear	7856	9116	9208

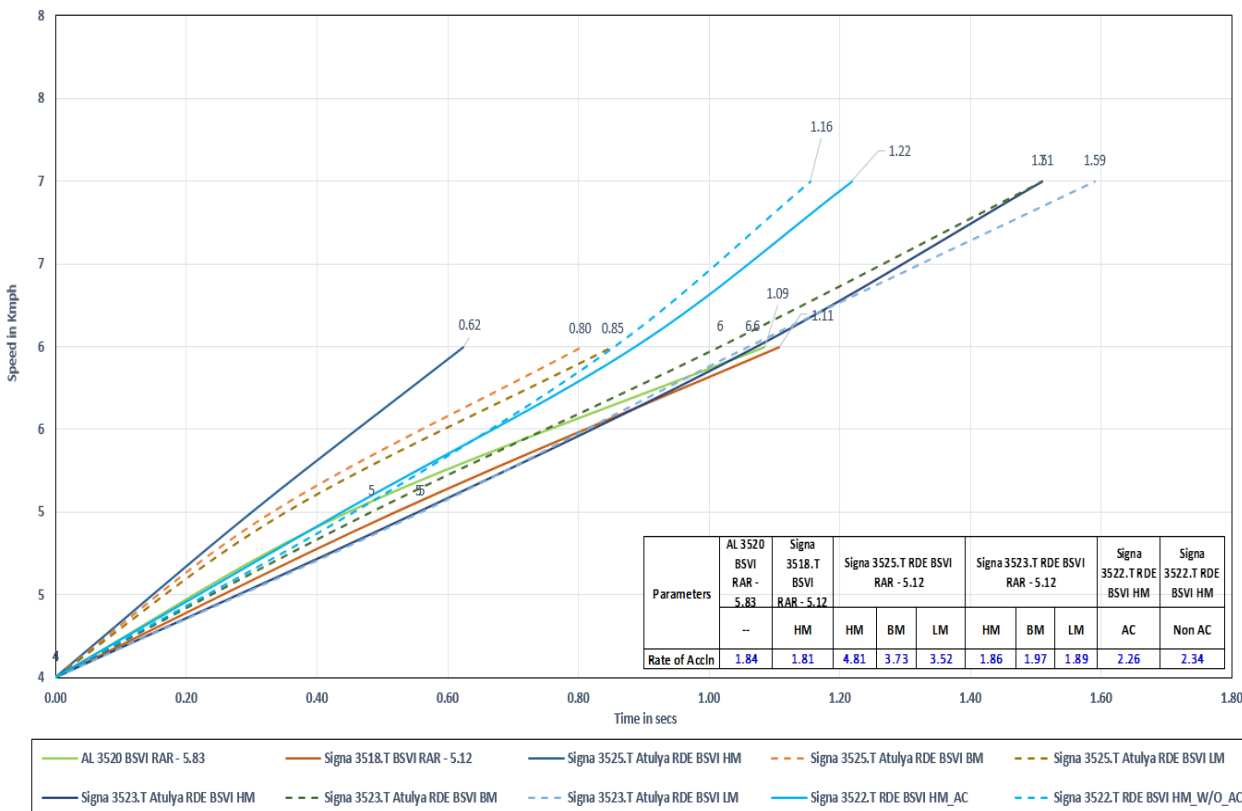
Acceleration Performance – Through Gear



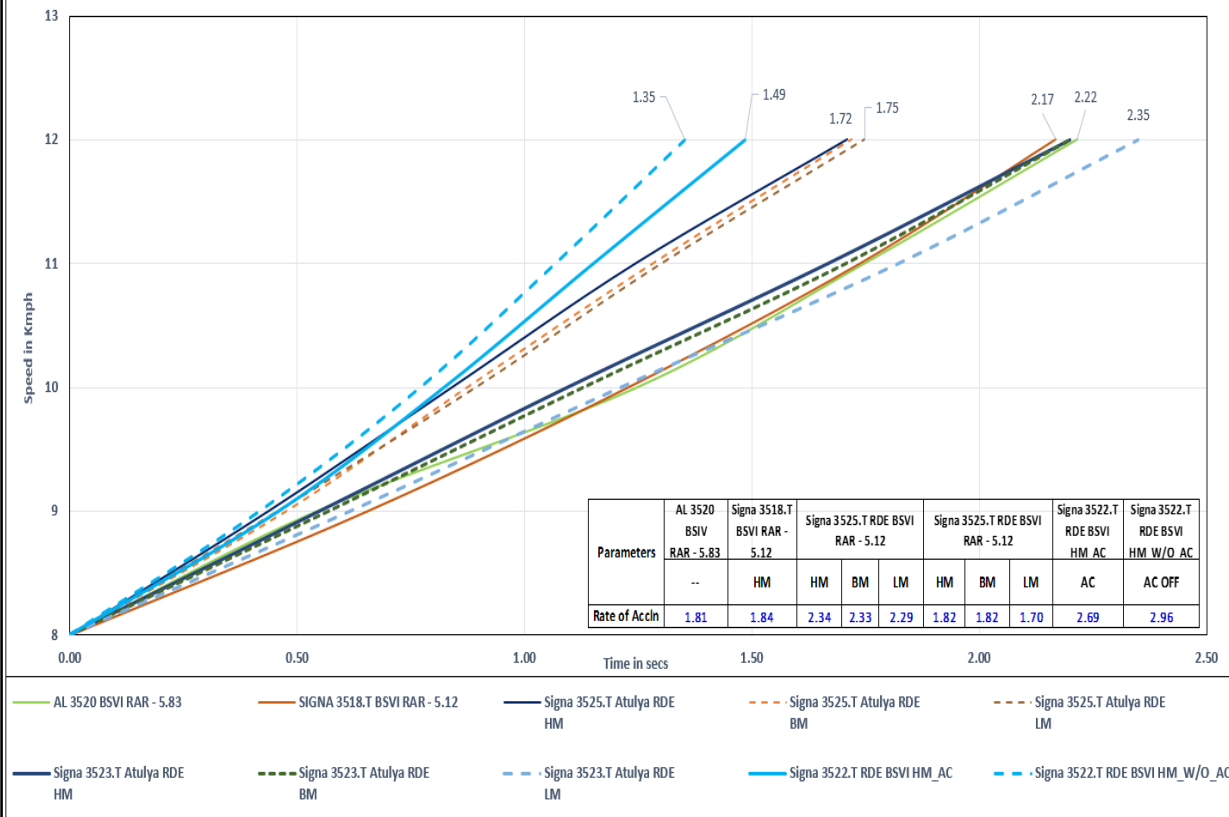
- ❖ Acceleration performance through gear for Signa 3522.T Cx BSVI RDE **is inferior than** AL 3520.T and Signa 3523.T BSVI RDE in 0 – 60 Kmph
- ❖ Acceleration performance through gear for Signa 3522.T Cx BSVI RDE **is inferior than** AL 3520.T and Signa 3523.T BSVI in 0 – 1000 meters

WOT Drivability @ 1st & 2nd Gear

Drivability @ 1st Gear

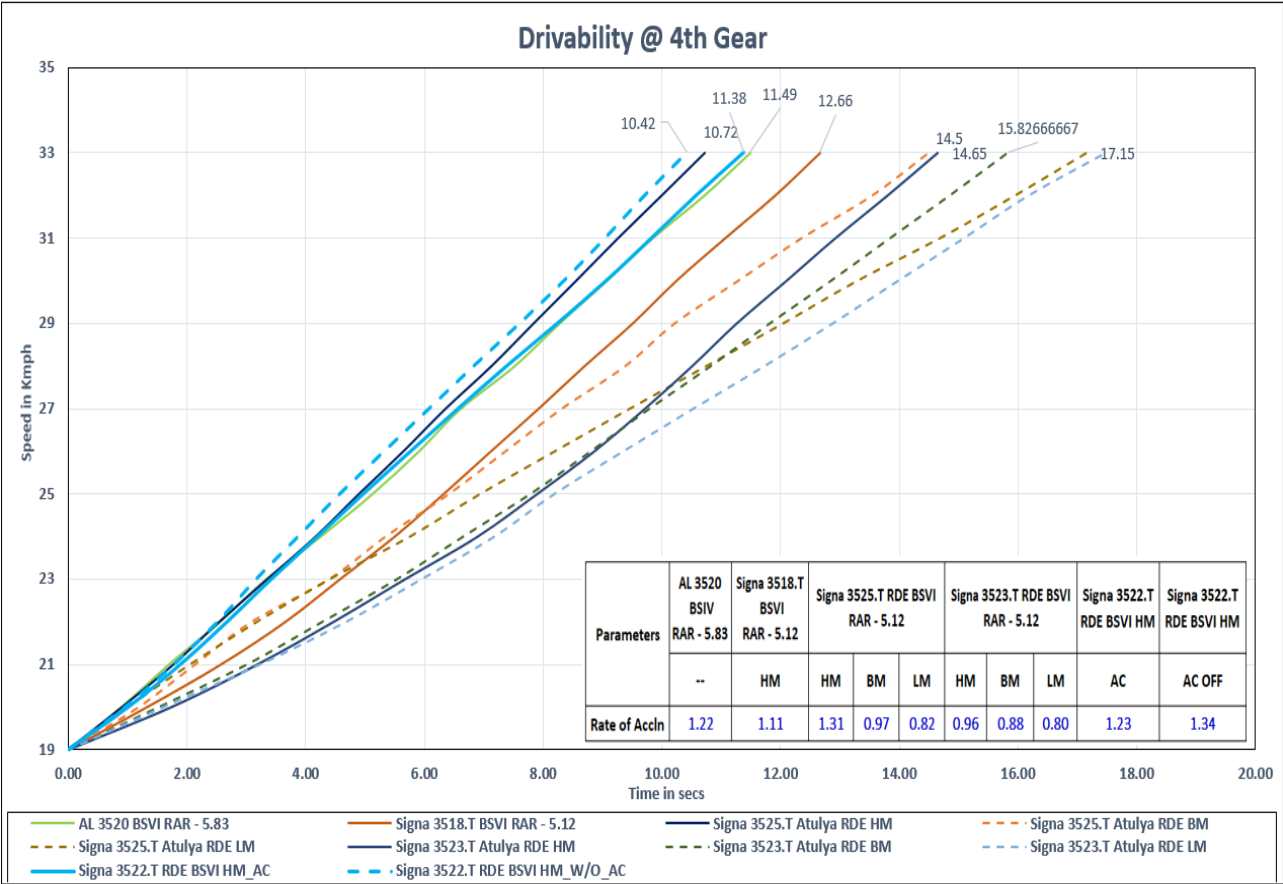
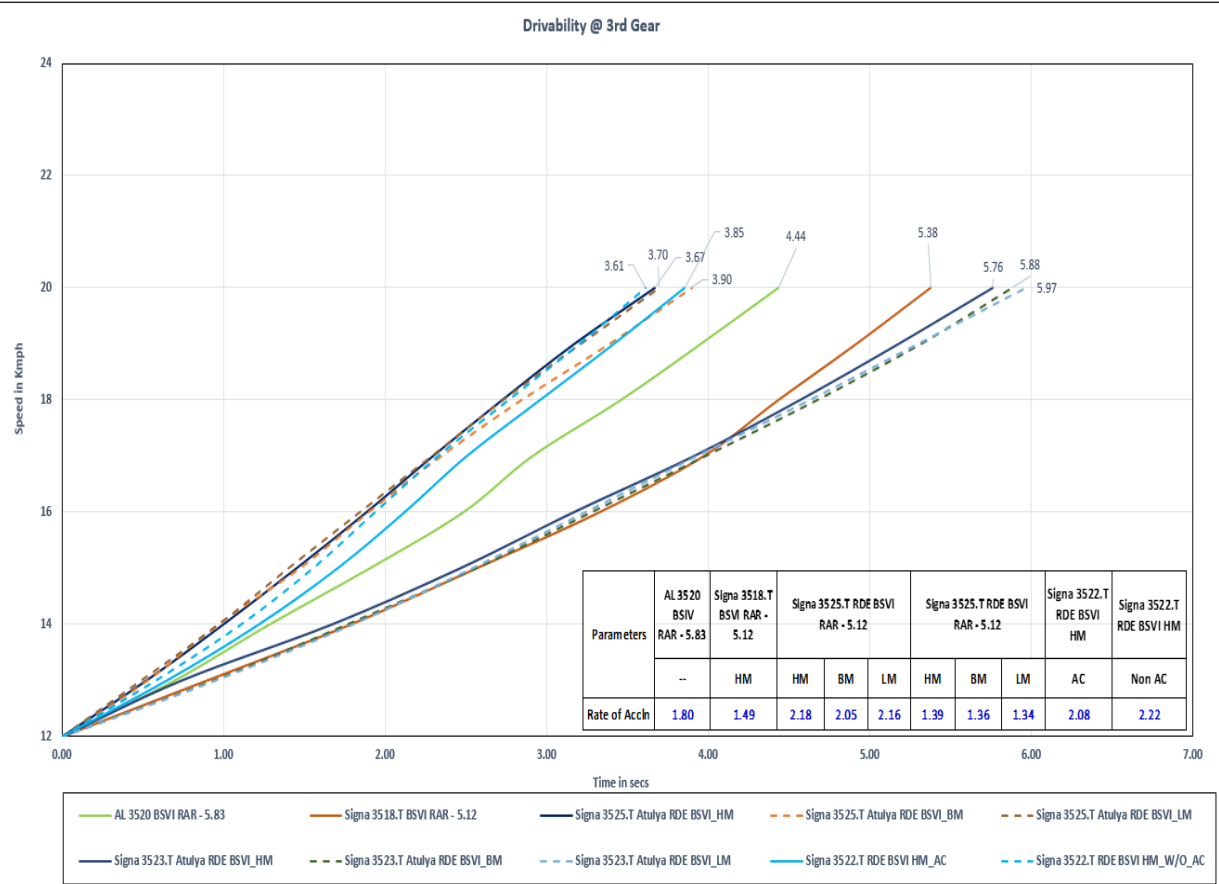


Drivability @ 2nd Gear



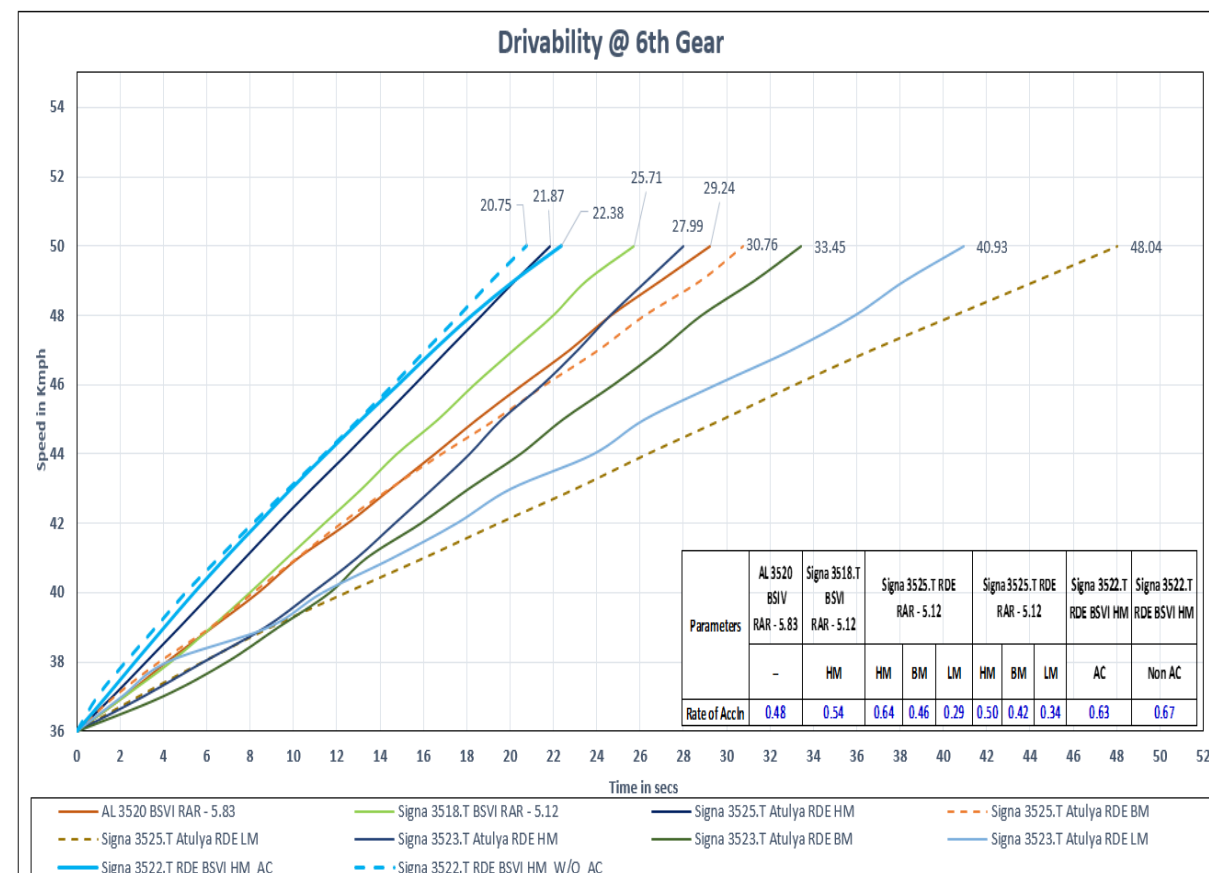
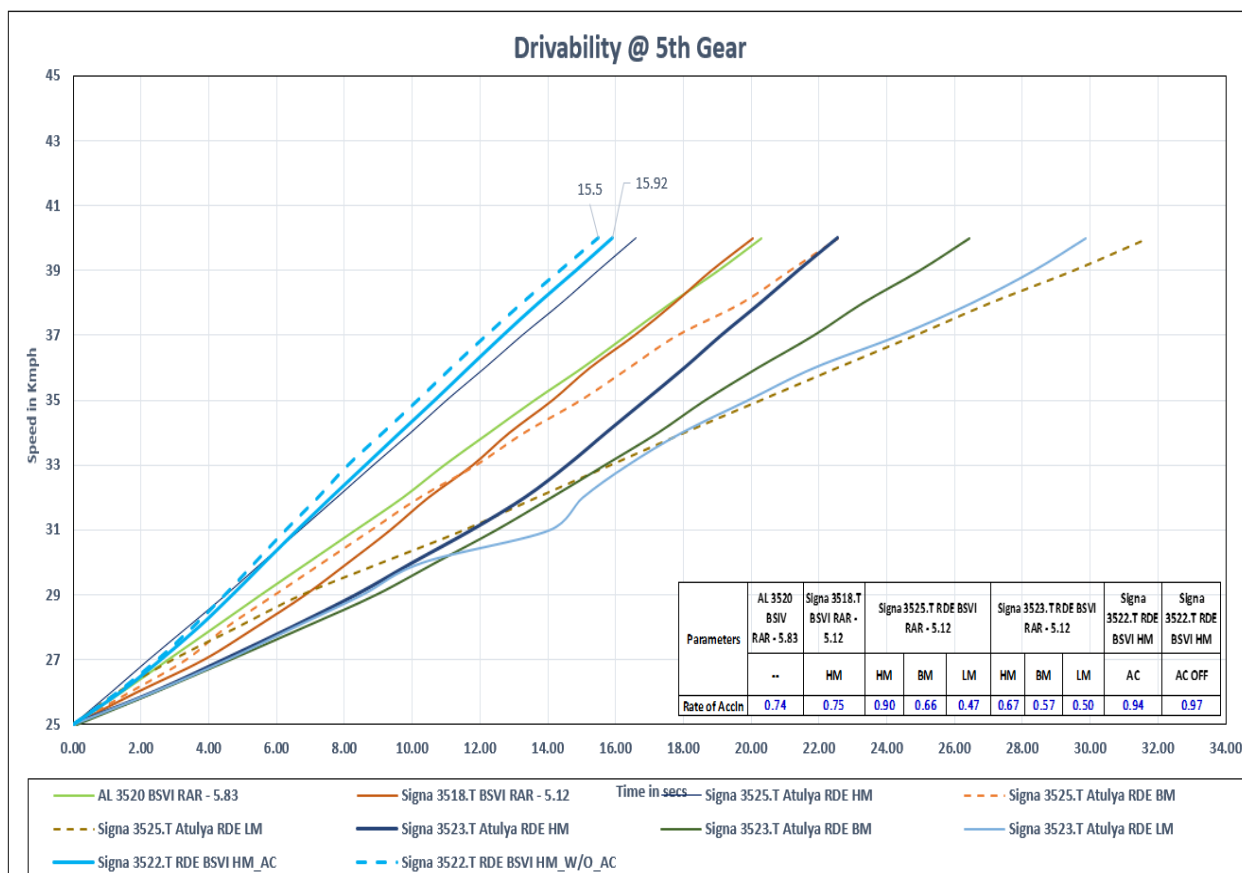
- ❖ In First gear, WOT acceleration performance of Signa 3522.T Cx RDE vehicle **is better as compared** with AL 3520.T BSVI and Signa 3523.T BSVI RDE
- ❖ In Second gear, WOT acceleration performance of Signa 3522.T Cx RDE vehicle **is better as compared** with AL 3520.T BSVI and at par Signa 3523.T BSVI RDE

WOT Drivability @ 3rd & 4th Gear



- ❖ In third gear, WOT acceleration performance of Signa 3522.T Cx RDE vehicle **is better as compared** with AL 3520.T BSVI and Signa 3523.T BSVI
- ❖ In fourth gear, WOT acceleration performance of Signa 3522.T Cx RDE vehicle **is better as compared** with AL 3520.T BSVI and Signa 3523.T BSVI

WOT Drivability @ 5th & 6th Gear

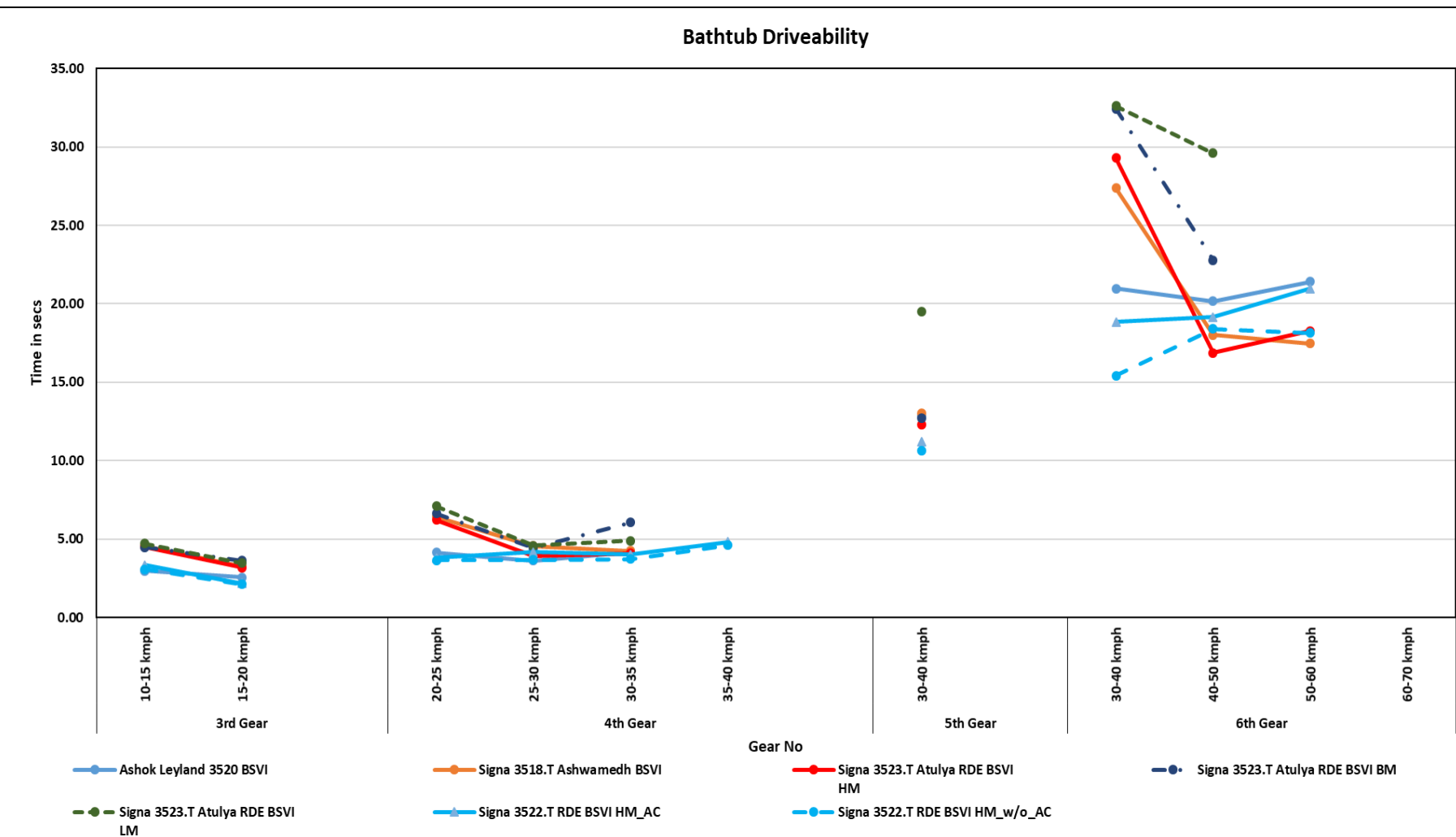


- ❖ In fifth gear, WOT acceleration performance of Signa 3522.T Cx RDE vehicle **is better as compared** with AL 3520.T BSVI and Signa 3523.T BSVI RDE
- ❖ In sixth gear, WOT acceleration performance of Signa 3522.T Cx RDE vehicle **is better as compared** with AL 3520.T BSVI and Signa 3523.T BSVI

Bathtub Drivability Report

Speed band	Ashok Leyland 3520 BSVI	Signa 3518.T Ashwamedh BSVI	Signa 3523.T Atulya RDE BSVI HM	Signa 3523.T Atulya RDE BSVI BM	Signa 3523.T Atulya RDE BSVI LM	Signa 3522.T RDE BSVI HM_AC	Signa 3522.T RDE BSVI HM_w/o_AC
	Time in Sec	Time in Sec	Time in Sec	Time in Sec	Time in Sec	Time in Sec	Time in Sec
10-15 kmph	2.96	4.55	4.51	4.47	4.71	3.32	3.07
15-20 kmph	2.55	3.26	3.19	3.63	3.46	2.17	2.12
20-25 kmph	4.13	6.43	6.20	6.64	7.09	3.80	3.64
25-30 kmph	3.63	4.55	3.94	4.42	4.57	4.16	3.67
30-35 kmph	4.20	4.23	4.09	6.07	4.89	4.01	3.71
35-40 kmph						4.81	4.61
30-40 kmph	12.85	13.01	12.29	12.7	19.50	11.19	10.64
30-40 kmph	20.97	27.37	29.30	32.42	32.60	18.84	15.41
40-50 kmph	20.15	18.00	16.88	22.77	29.60	19.16	18.40
50-60 kmph	21.41	17.46	18.26			20.96	18.14
60-70 kmph							

Bathtub Drivability Comparson with Benchmark & Base Vehicles



Remarks:.

Vehicle is meeting PALS target in all gears except 3rd Gear 10 – 15 Kmph and 4th gear 20 – 25 and 25 - 30 kmph

Thank You!

Object: Testing-Outdoor-UA/01941/A;1

Revision: A;1

Owner:  Patil Shubham (s0001158.ttl) ▼Group ID:  ERC_Designers_Group ▼Last Modifying User:  Ajaykumar Jadhav (ahj654087) ▼

Request Domain: Testing-Outdoor ▼

Project Code: 5201 ▼

Project Description: Request from Mr.Amit Gupta, Mr.Amit Kumar Dhayade,
Dated 14.09.2022

Buisness Unit: CVBU

Work Breakdown Structure (WBS): P.23.EA.504 ▼

WBS Description: 5.3L CNG RDE Engine on M&HCV Truck Segme

Testing Sub Group: PD-Performance and Driveability ▼

Aggregates: NA ▼

Program: MHCV-T3 ▼

Design Status: DR3 ▼

Drawing No: 52010160R

Test Request Reference No: -

Ref DVP No:

Request Agency: Vehicle-I (Cvbu) ▼

Vehicle/System Stage(mule,alpha,Beta,PO,PP,etc): Beta ▼

Supplier: TML

Investigation Type/Reason for Test: New development ▼

TATA MOTORS LIMITED ERC PUNE	TEST PLAN	DATE: 20/08/2025			
	VEHICLE TESTING DEPARTMENT				
Description: Signa 3522.T CNG RDE AC PD evaluation		Test Item No: UA001816			
		New	✓	Moderate	✗
		Major	✗	Minor	✗

Part No: 52010160R, Drawing No: 52010160R	Modification No: G;2,	Enclosures: Nil
Make: TML	WBS Number: P.23.EA.504- 5.3L CNG RDE Engine on M&HCV Truck Segme	Copies to :
Vehicle Type: Request from Mr.Amit Gupta, Mr.Amit Kumar Dhayade, Dated 14.09.2022	No. of Samples: -	
Test Requested by:		Equipments: Nil
Test Requisition Reference:	Testing-Outdoor-UA/01941	
Reason for Test/s:	New development	
Regulatory Requirement:	a) Component: NA	
	b) Vehicle: NA	

REMARKS: Test plan for DVP Execution	INTERNAL REF:	NA
TEST TO FAILURE: No		
MEOST: No		
RELIABILITY TESTING: No		
BENCHMARKING TESTING: No		
DFMEA/FMEA DONE: Yes		
PREPARED BY	CHECKED BY	APPROVED BY
Amit Satapathy		